Software Requirements Specification

hospital

Req2Design · IEEE 830-1998 aligned

Document Control

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1.0	2026-04-01	abdulhaq	Initial generated draft for structured review.

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1.

1Introduction

1.1PurposeThe purpose of this document is to specify the functional and non-functional requirements for a Hospital Management System that facilitates patient-doctor interactions, appointment scheduling, and administrative tasks.

1.2ScopeThis SRS covers the development of a web-based application that manages patient-doctor interactions, appointment scheduling, and administrative tasks related to patient care.

1.3Definitions/Acronyms- HMS:Hospital Management System

- **SMS:** Short Message Service
- **API:** Application Programming Interface
- **UI:** User Interface

1.4References- IEEE Standard for Software Requirements Specifications, IEEE Std 830-1998

- Hospital Management System Design Document

1.5OverviewThis document outlines the requirements for a system that enables patients to find doctors by specialty, book appointments, and receive reminders. It also provides functionalities for doctors to manage their schedules and patient visits. ---

2.

2Overall Description

2.1Product PerspectiveThe product perspective includes the scope of the system, its intended users, constraints, and assumptions.

2.2Product FunctionsThe system will provide the following functions:

- Patient can search for doctors by specialty and book appointments.
- Patients receive SMS or email reminders for upcoming appointments.
- Doctors can block unavailable time slots, view daily schedules, and mark completed visits.
- The system handles walk-ins, reschedules, cancels appointments, and prevents double bookings.
- Patient basic history is visible during booking.
- The system maintains a log of booking changes for staff review, with a maximum size of 200 MB.

2.3User Characteristics- Patients:Individuals seeking medical care who can use the system to book appointments and receive reminders.

- **Doctors:** Medical professionals who can manage their schedules and patient visits.

2.4Constraints- Memory usage shall not exceed 200 MB.

- The system must support SMS and email notifications.
- The system must be accessible via web browsers.

2.5Assumptions/Dependencies- The system will be hosted on a reliable server infrastructure.

- Users have access to internet-enabled devices.
- The system will integrate with existing hospital databases for patient history. ---

3.

3Specific Requirements

3.1External Interface Requirements- User Interface (UI):- Patient UI:A web interface allowing patients to search for doctors by specialty, book appointments, and receive reminders.

- **Doctor UI:** A web interface allowing doctors to block unavailable time slots, view daily schedules, and mark completed visits.
- **Hardware Interface:** - The system will run on standard web servers and client devices with modern web browsers.
- **Software Interface:** - Integration with hospital databases for patient history.
- Integration with SMS and email services for sending reminders.
- **Communication Interface:** - The system will use HTTP/HTTPS for communication between the client and server.
- SMS and email services for sending reminders.

3.2Functional Requirements-

FR-1 Doctor Search and Appointment Booking-

Input: Patient searches for doctors by specialty.

Processing: System retrieves available doctors and their schedules.

Output: Display list of available doctors and allow patient to select a doctor and book an appointment.

Priority: High

- **FR-2:** Appointment Reminders-

Input: Appointment details.

Processing: System sends SMS or email reminders to the patient before the scheduled appointment.

Output: Notification sent to the patient.

Priority: High

- **FR-3:** Doctor Schedule Management-

Input: Doctor logs into the system.

Processing: System displays the doctor's daily schedule.

Output: Doctor can block unavailable time slots and mark completed visits.

Priority: High

- **FR-4:** Walk-In Handling-

Input: Patient arrives at the clinic without a booked appointment.

Processing: System allows the doctor to accept the walk-in appointment.

Output: Appointment is added to the doctor's schedule.

Priority: Medium

- **FR-5:** Rescheduling and Cancellation-

Input: Patient or doctor requests to reschedule or cancel an appointment.

Processing: System updates the appointment status and notifies all relevant parties.

Output: Updated appointment status and notifications sent.

Priority: Medium

- **FR-6:** Prevent Double Booking-

Input: Attempt to book an already occupied time slot.

Processing: System checks the availability and prevents double booking.

Output: Error message displayed to the patient.

Priority: High

- **FR-7:** Patient History Visibility-

Input: Patient books an appointment.

Processing: System retrieves the patient’s basic history from the database.

Output: Patient’s basic history is displayed to the doctor.

Priority: Medium

3.3Non-Functional Requirements- Performance:- The system must respond within 2 seconds to user requests.

- The system must handle up to 100 concurrent users without degradation.
- **Security:** - All data transmitted between the client and server must be encrypted using HTTPS.
- User passwords must be stored securely using hashing algorithms.
- **Reliability:** - The system must be available 99.9% of the time.
- The system must recover from failures within 1 hour.
- **Usability:** - The system must be easy to use and navigate.
- The system must provide clear error messages and help documentation.
- **Maintainability:** - The system must be modular and easily upgradable.
- The codebase must be well-documented and follow best practices.
- **Portability:** - The system must be deployable on multiple platforms and web servers.

3.4

System Features-

Feature 1: Doctor Search and Appointment Booking- Patients can search for doctors by specialty and book appointments.

- Patients receive SMS or email reminders for upcoming appointments.
- **Feature 2:** Doctor Schedule Management- Doctors can block unavailable time slots and view their daily schedules.
- Doctors can mark completed visits.
- **Feature 3:** Walk-In Handling- The system allows doctors to accept walk-in appointments.
- **Feature 4:** Rescheduling and Cancellation- Patients and doctors can reschedule or cancel appointments.
- **Feature 5:** Prevent Double Booking- The system prevents double booking by checking the availability of time slots.
- **Feature 6:** Patient History Visibility- Patient basic history is visible during booking. --- End of Document.